



Ordinary Differential Equations

First-Order Differential Equations

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ENGR 213**

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Today's Lecture

- Separable Equations
- Linear Equations

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 2.2, 2.3**

Separable Equations

□ A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is said to be **separable** or to have **separable variables**.

e.g.: $\frac{dy}{dx} = x^2y^4e^{5x-3y}$

$$\frac{dy}{dx} = y + \cos(x)$$

□ Method of solution

$$\frac{dy}{dx} = g(x)h(y) \longrightarrow \overset{p(y)}{\frac{1}{h(y)}\frac{dy}{dx} = g(x)} \longrightarrow p(y)dy = g(x)dx$$

$$p(y)dy = g(x)dx \longrightarrow \int p(y)dy = \int g(x)dx \longrightarrow H(y) = G(x) + c$$


 $H(y)$ and $G(x)$ are antiderivatives
of $p(y)$ and $g(x)$

EXAMPLE 1

Solve $(1 + x)dy - ydx = 0$.

EXAMPLE 2

Solve the initial-value problem $\frac{dy}{dx} = \frac{-x}{y}$, $y(4) = -3$.

EXAMPLE 3

Solve $\frac{dy}{dx} = y^2 - 4$.



Linear Equations

- A first-order differential equation of the form

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

is a **linear equation** in the dependent variable y .

- When $g(x) = 0$, this linear equation is said to be **homogeneous**; otherwise, it is **nonhomogeneous**.
- The **standard form** of a linear DE is obtained by dividing both sides by the lead coefficient ($a_1(x)$).

$$\frac{dy}{dx} + P(x)y = f(x)$$

□ The standard form equation has the property that its solution y is the sum of the two solutions $\rightarrow y = y_c + y_p$

- $\mathbf{y_c}$ is a solution of the associated homogeneous equation:

$$\frac{dy}{dx} + P(x)y = 0 \xrightarrow{\text{separable}} \frac{dy}{y} + P(x)dx = 0$$

$$\frac{dy}{y} + P(x)dx = 0 \xrightarrow{\text{solving for } y} y_c = ce^{-\int P(x)dx} \text{ or } y_c = cy_1(x)$$

- $\mathbf{y_p}$ is a particular solution of the nonhomogeneous equation:

$$\frac{dy_p}{dx} + P(x)y_p = f(x) \xrightarrow{y_p = u(x)y_1(x)} y_p(x) = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x)dx$$

integrating factor

❑ Method of solution

Steps:

- I. Put a linear first-order equation into standard form and determine $P(x)$ and the **integrating factor**.

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \longrightarrow \frac{dy}{dx} + P(x)y = f(x) \quad \begin{cases} P(x) \\ e^{\int P(x)dx} \end{cases}$$

- II. Multiply the standard form by the integrating factor.

$$\underbrace{e^{\int P(x)dx} \frac{dy}{dx} + e^{\int P(x)dx} P(x)y}_{\frac{d}{dx} [e^{\int P(x)dx} y]} = e^{\int P(x)dx} f(x)$$

- III. Integrate both sides of the resulting DE.



EXAMPLE 4

Solve $\frac{dy}{dx} - 3y = 6$.

EXAMPLE 5

Solve $(x^2 - 9) \frac{dy}{dx} + xy = 0$.

□ Piecewise-linear differential equation

- In the construction of mathematical models (especially in the biological sciences and engineering) it can happen that one or more coefficients in a differential equation is a piecewise-defined function.
- The idea is to solve the IVP in parts corresponding to the intervals over which function(s) is defined and then piece the solutions together.

EXAMPLE 5

Solve $\frac{dy}{dx} + y = f(x), y(0) = 0$

where $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$



